

Model performance can also be adversely affected by data drift even when the deployment settings are the same.

Although GitOps controllers take care of achieving the desired settings of the system, separate monitoring is needed to assess the prediction performance of the model and its fairness, latency and evolving data patterns.

Security has become a prime concern. Git repositories can contain sensitive configuration data, access credentials, or infrastructure information. Secrets should not be stored in plain-text format. The best way to safeguard secrets is by using secret management systems combined with role-based access. The approval process for certain high-risk production deployments should also be automated.

Several best practices can improve implementation:

- Maintain separate repositories

or clearly defined directories for application code, ML pipelines and deployment configurations.

- Assign unique versions to datasets, models, container images and infrastructure configurations.
- Define measurable acceptance criteria before promoting a model to production.
- Use pulls requests and peer reviews for production-related changes.
- Implement continuous monitoring for data drift, model drift, latency and resource usage.
- Use canary or gradual deployment strategies to reduce the risk of model updates.
- Maintain tested rollback procedures for restoring a stable model quickly.

By combining these practices with appropriate governance, organisations can build AI deployment systems that are automated, secure, traceable and reliable.

The application of GitOps in AI will involve the increased use of automation, scalability and governance in model management in the future. GitOps platforms will have the ability to detect data drift or model drift autonomously, retrain models, assess models, and eventually implement them once particular performance, safety, and ethical standards are met. Moreover, the GitOps model will be needed for the consistent implementation of models in all the cloud, hybrid and edge scenarios. Due to enhanced governance and monitoring based on policies, organisations will be able to connect every model action to its data, code, configuration and deployment log. **END** 🐧

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